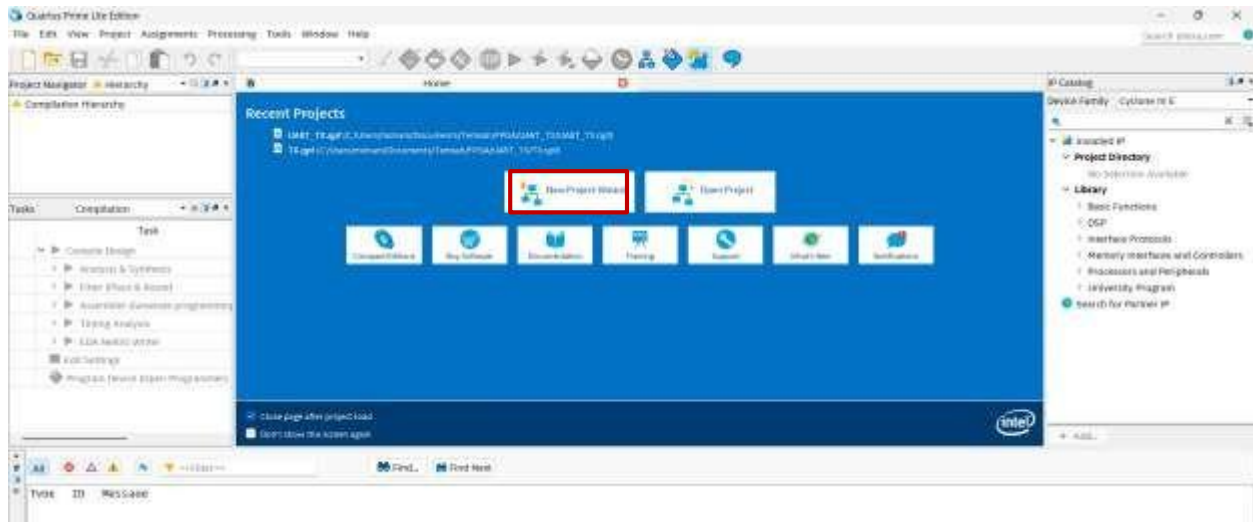


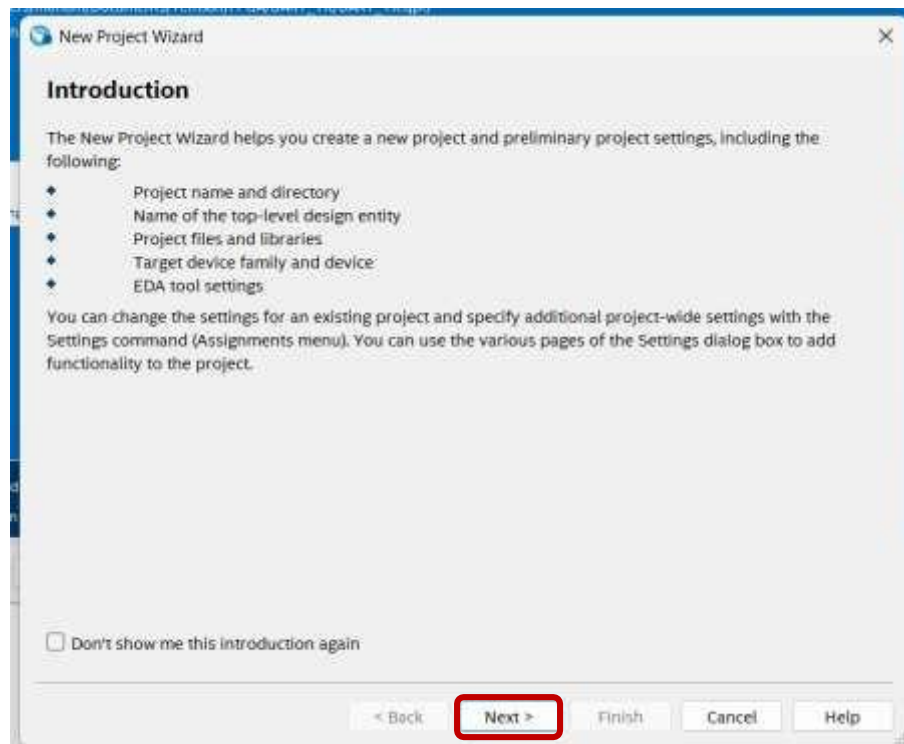
Lab 2.0

Step 1: Project setup

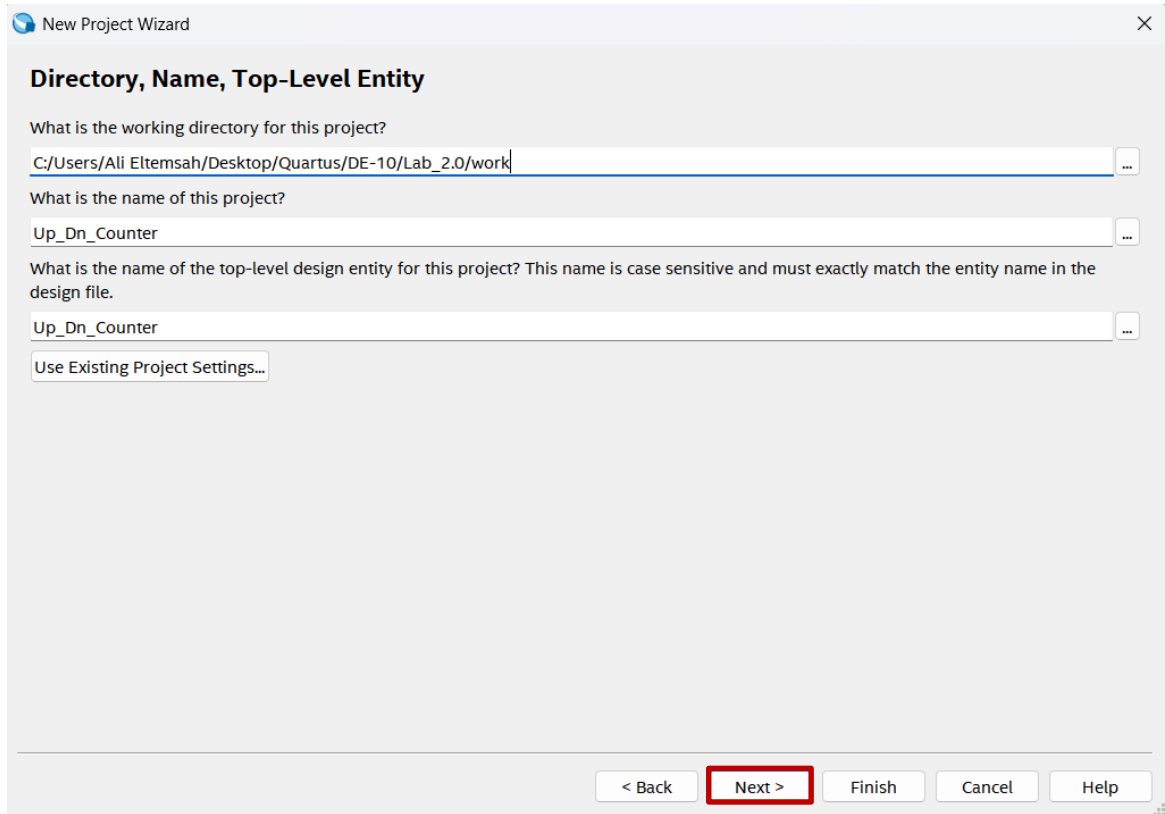
- Launch **Intel Quartus Prime Lite Edition** and click on **New Project Wizard** on the home landing page.
- Initializes a new design workspace, where the project directory, top-level entity name (UpDnCounter), target FPGA family (Cyclone IV), and EDA tools are defined.



- Click **Next >** on the wizard introduction dialog to proceed with project configuration.

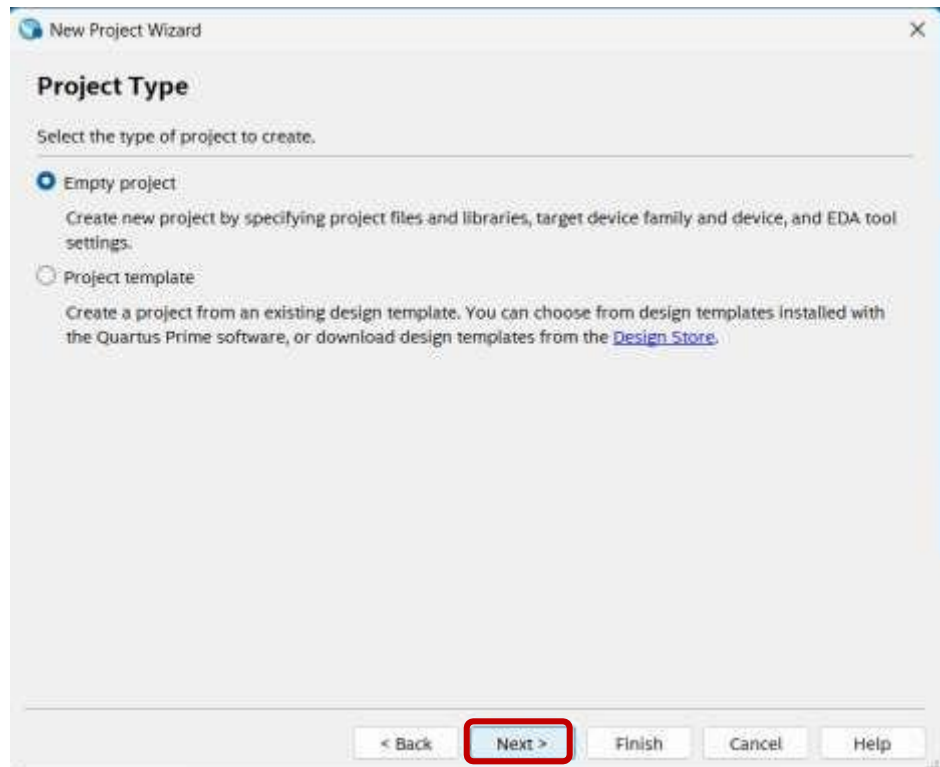


- Set the working directory path, name the project, and set the top-level entity name to Up_Dn_Counter. Click **Next >**.



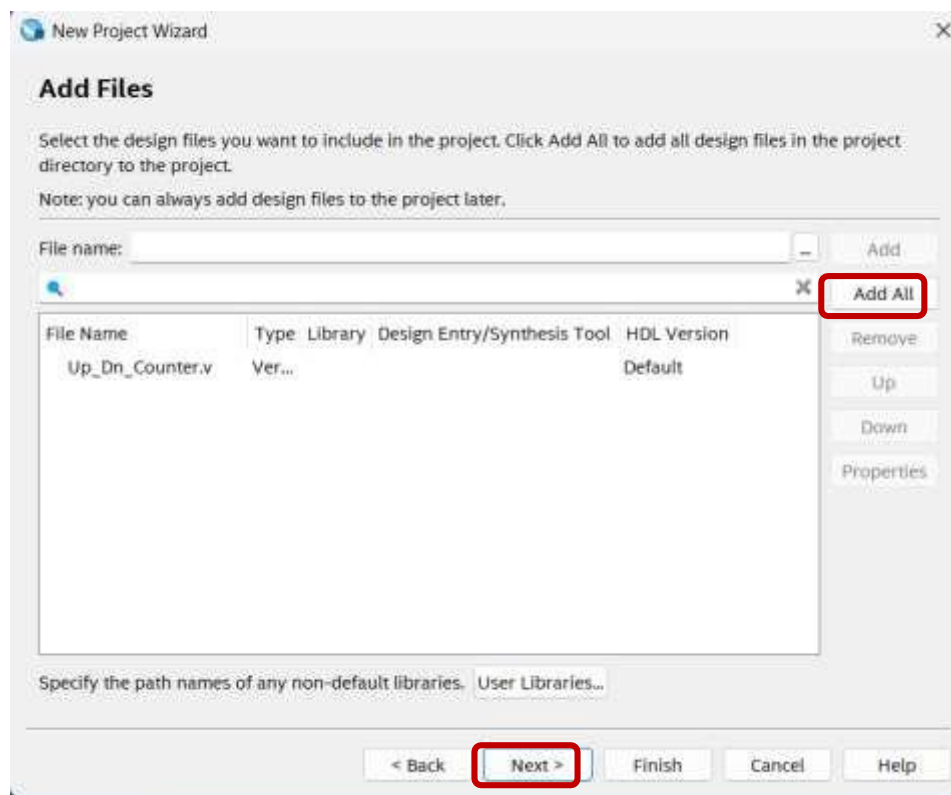
The screenshot shows the 'New Project Wizard' dialog box with the title 'Directory, Name, Top-Level Entity'. It contains three text input fields: 'What is the working directory for this project?' with the path 'C:/Users/Ali Eltemsah/Desktop/Quartus/DE-10/Lab_2.0/work', 'What is the name of this project?' with 'Up_Dn_Counter', and 'What is the name of the top-level design entity for this project?' with 'Up_Dn_Counter'. There is a 'Use Existing Project Settings...' button. At the bottom, there are five buttons: '< Back', 'Next >' (highlighted with a red box), 'Finish', 'Cancel', and 'Help'.

- Select **Empty project** and click **Next >** to create a fresh workspace for importing HDL source files and IP blocks.

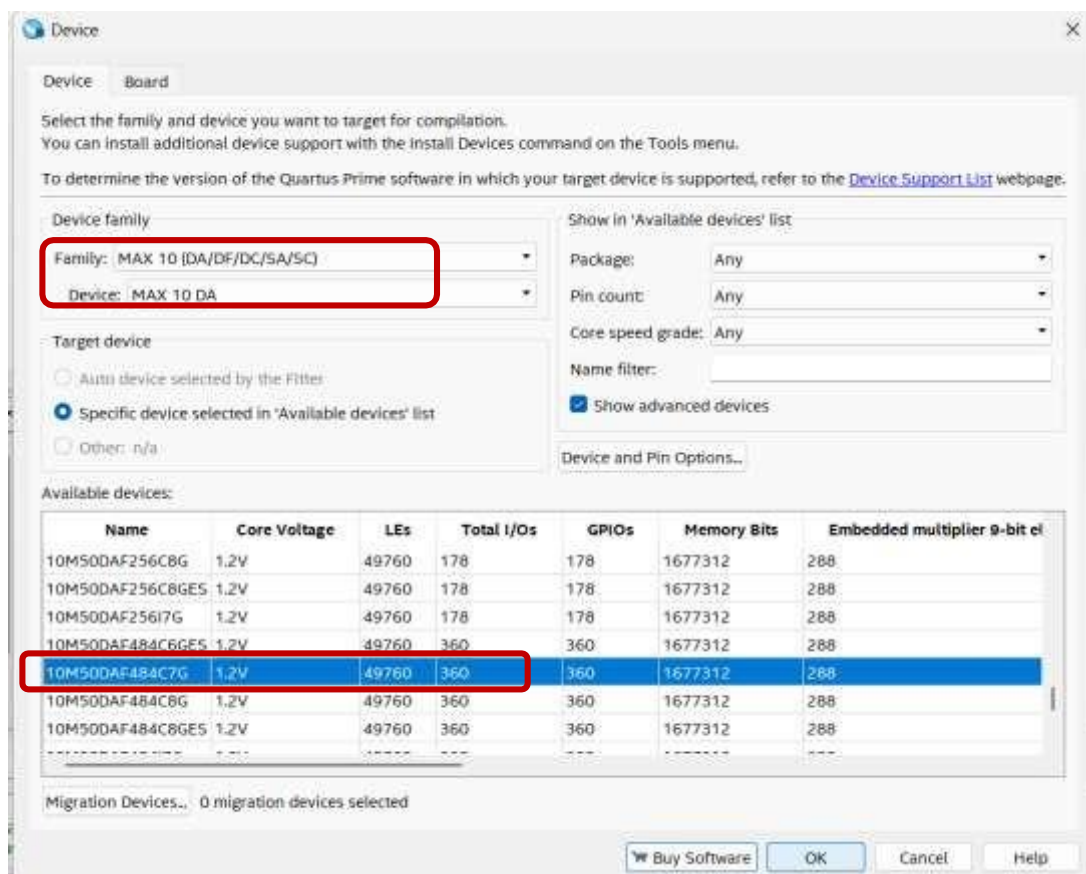


The screenshot shows the 'New Project Wizard' dialog box with the title 'Project Type'. It contains two radio button options: 'Empty project' (selected) and 'Project template'. The 'Empty project' option has a description: 'Create new project by specifying project files and libraries, target device family and device, and EDA tool settings.' The 'Project template' option has a description: 'Create a project from an existing design template. You can choose from design templates installed with the Quartus Prime software, or download design templates from the [Design Store](#).' At the bottom, there are five buttons: '< Back', 'Next >' (highlighted with a red box), 'Finish', 'Cancel', and 'Help'.

- Add the Verilog design to the project by clicking **Add All**, then click **Next >**.

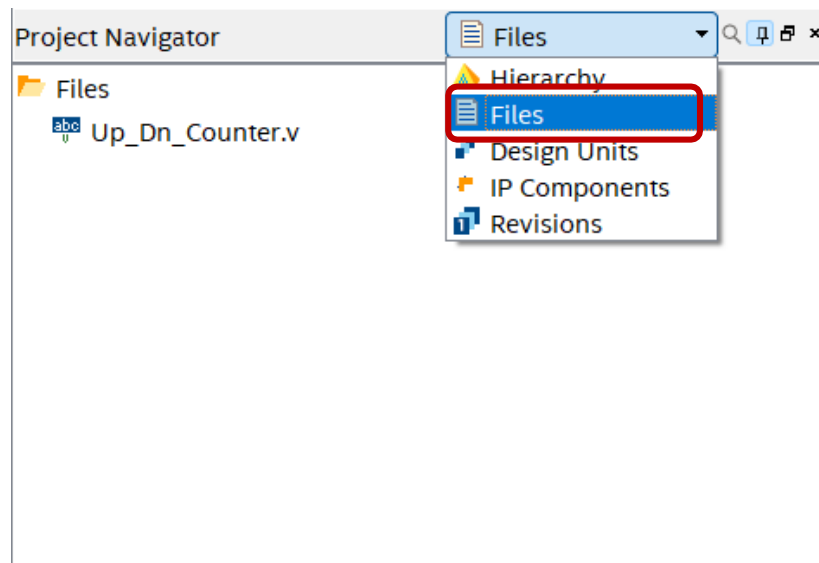


- Set the Device family to **MAX 10**, select the target FPGA device **10M50DAF484C7G**, and click **Finish** to instantiate the workspace.

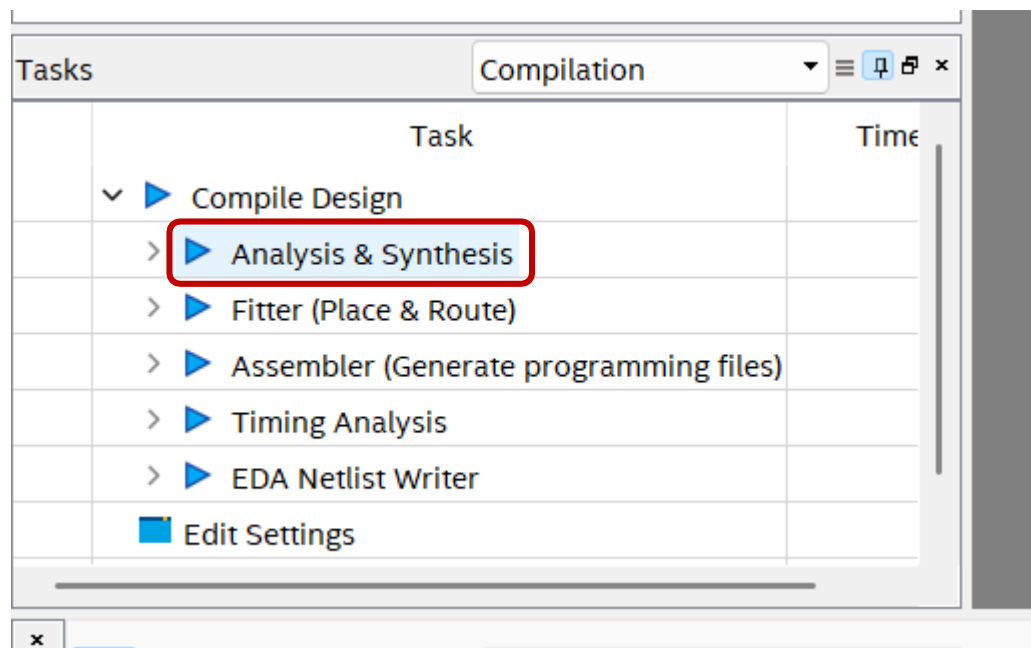


Step 2: Design Entry

- Switch the Project Navigator dropdown view from **Hierarchy** to **Files** to display all added design source files.



- In the Tasks pane on the left, double-click **Analysis & Synthesis** to verify HDL syntax and synthesize the netlist into device logic elements.



Step3: I/O and Pin configuration

- Open **Pin Planner** by clicking the Pin Planner icon on the top toolbar

File Edit View Processing Tools Window Help

Report

Report not available

Groups

Report

Tasks

Early Pin Planning

Early Pin Planning...

Run I/O Assignment /

Export Pin Assignment

Pin Finder...

Highlight Pins

I/O Banks

Named: *

Edit: X

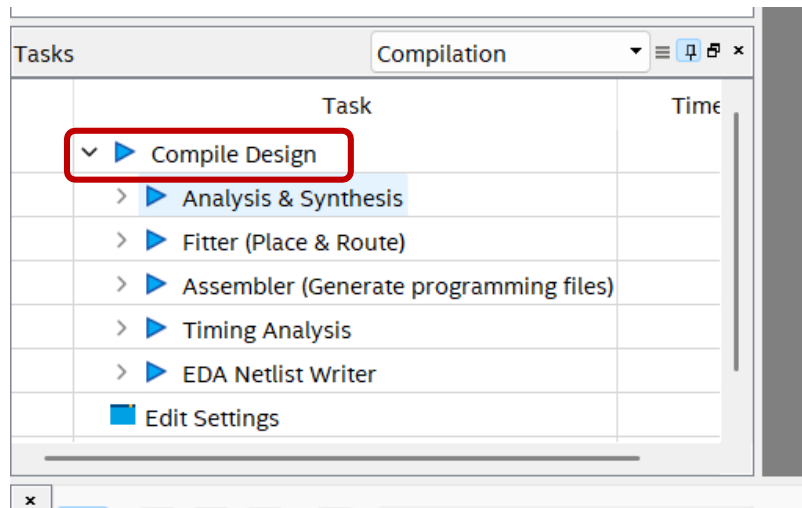
Node Name	Direction	Location	I/O Bank	VREF Group	Pin Location	I/O Standard	Reserved	Current Strength	Slew Rate	Differential Pair	Input Preservation
Counter[4]	Output	PIN_D13	7	B7_NO	PIN_D13	2.5 V		12mA ..ault)	2 (default)		
Counter[3]	Output	PIN_B10	7	B7_NO	PIN_B10	2.5 V		12mA ..ault)	2 (default)		
Counter[2]	Output	PIN_A10	7	B7_NO	PIN_A10	2.5 V		12mA ..ault)	2 (default)		
Counter[1]	Output	PIN_A9	7	B7_NO	PIN_A9	2.5 V		12mA ..ault)	2 (default)		
Counter[0]	Output	PIN_A8	7	B7_NO	PIN_A8	2.5 V		12mA ..ault)	2 (default)		
Down	Input	PIN_A13	7	B7_NO	PIN_A13	2.5 V		12mA ..ault)			
High	Output	PIN_B11	7	B7_NO	PIN_B11	2.5 V		12mA ..ault)	2 (default)		
IN[4]	Input	PIN_A12	7	B7_NO	PIN_A12	2.5 V		12mA ..ault)			
IN[3]	Input	PIN_C12	7	B7_NO	PIN_C12	2.5 V		12mA ..ault)			
IN[2]	Input	PIN_D12	7	B7_NO	PIN_D12	2.5 V		12mA ..ault)			
IN[1]	Input	PIN_C11	7	B7_NO	PIN_C11	2.5 V		12mA ..ault)			
IN[0]	Input	PIN_C10	7	B7_NO	PIN_C10	2.5 V		12mA ..ault)			
Load	Input	PIN_B12	7	B7_NO	PIN_B12	2.5 V		12mA ..ault)			
Low	Output	PIN_A11	7	B7_NO	PIN_A11	2.5 V		12mA ..ault)	2 (default)		
Up	Input	PIN_A14	7	B7_NO	PIN_A14	2.5 V		12mA ..ault)			
clk	Input	PIN_P11	3	B3_NO	PIN_P11	2.5 V		12mA ..ault)			
rst	Input	PIN_F15	7	B7_NO	PIN_F15	2.5 V		12mA ..ault)			
<<new node>>											

Top View - Wire Bond

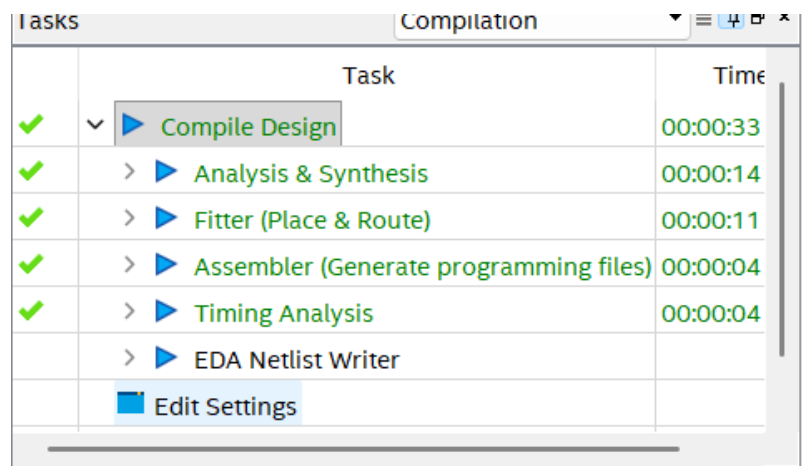
MAX 10 - 10M50DAF484C7G

Step 4: Compilation Flow

- Double-click **Compile Design** in the Tasks pane to run the complete compilation flow, including Analysis & Synthesis, Fitter (Place & Route), Assembler, and Timing Analysis.



- Verify green checkmarks across all compilation stages



Step 5: Program Device

Double-click **Program Device** in the Tasks pane. **Programmer Window** will appear. Check the hardware setup and then Press start

	Task	Time
✓	> ▶ Fitter (Place & Route)	00:00:57
✓	> ▶ Assembler (Generate programming files)	00:00:57
	> ▶ Timing Analysis	
	> ▶ EDA Netlist Writer	
	Edit Settings	
	▶ Program Device (Open Programmer)	

Programmer - C:/Users/Ali Eltemsah/Desktop/Quartus/DE-10/Lab_2.0/work/Up_Dn_Co...

File Edit View Processing Tools Window Help Search altera.com

Hardware Setup... USB-Blaster [USB-0] Mode: JTAG Progress: % (Success)

☒ Enable real-time ISP to allow background programming when available

Start Stop Auto Detect Delete Add File... Change File Save File Add Device Up Down

File	Device	Checksum	Usercode	Program/Configure	Verify	Blank-Check	Examine
output_files/Up_...	10M50DAF484	00275CC9	00275CC9	☒	☐	☐	☐

TDI

TDO

10M50DAF484

